ANOVA – Analysis of Variance

Exploratory Data Analysis · Hypothesis Testing Module

What We'll Cover

01

Definition & Purpose

What ANOVA is and why it matters

02

Mechanics

Between-group vs. within-group variation and the F-statistic

03

Hypothesis Testing

Formulating and interpreting hypotheses

04

Worked Example

Teaching methods and exam scores — step by step

> *"ANOVA allows us to ask a single, powerful question: are these group means genuinely different, or are the differences just noise?"*

ANOVA is one of the most widely applied inferential statistical techniques in data analysis, enabling researchers and analysts to rigorously compare outcomes across multiple groups with a single unified test.

This module is part of the Hypothesis Testing unit within the Exploratory Data Analysis course.

ANOVA: Definition & Purpose

Definition: ANOVA (Analysis of Variance) is a statistical hypothesis testing method used to determine whether there is a statistically significant difference between the means of **three or more independent groups**.

What does ANOVA stand for?

Analysis of Variance — despite the name, it ultimately compares **group means**, not variances directly.

Core function

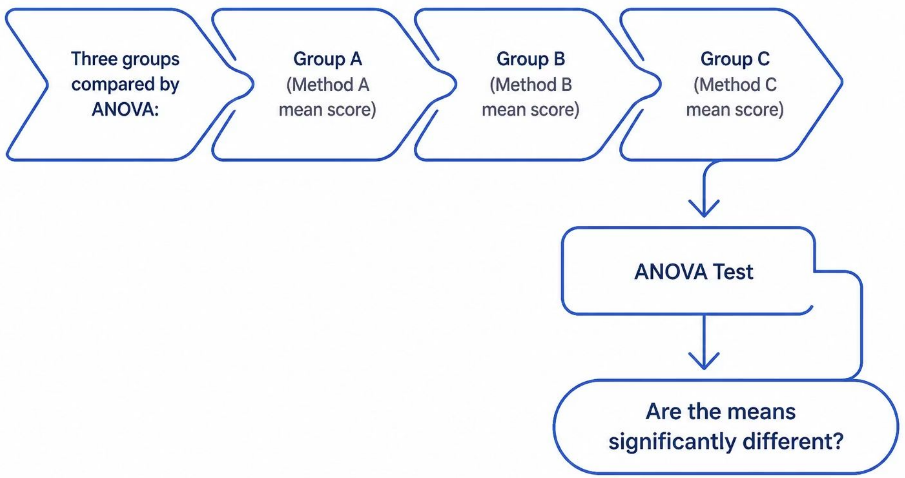
Determines whether observed differences between group means are **statistically significant** or simply due to random chance.

Role in EDA

Widely used to understand relationships between **categorical** (grouping) and **numerical** (outcome) variables.

Motivating Example

Comparing the average exam marks of students taught using **three different teaching methods** — a classic multi-group comparison problem.



ANOVA collapses the complexity of comparing all group pairs into a **single unified test**, controlling the overall Type I error rate.

How Does ANOVA Work?

ANOVA works by partitioning the **total variation** in the data into two components and comparing their magnitudes.

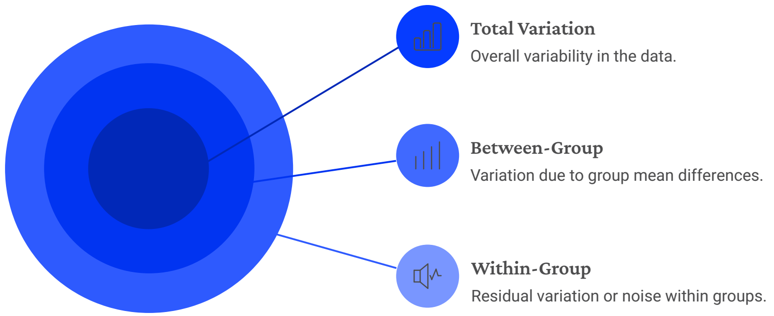
Between-Group Variation

Variation caused by **differences between the group means**. Reflects the effect of the grouping factor (e.g., teaching method). Large when groups are very different from each other.

Within-Group Variation

Variation among observations **inside the same group**. Reflects natural individual-level variability. Serves as the baseline "noise" in the data.

Total Variation Decomposition



The F-Statistic

The ratio at the heart of ANOVA:

$$F = \frac{\text{Between-Group Variance (MSB)}}{\text{Within-Group Variance (MSW)}}$$

Large F-value

Differences between group means are **large** relative to within-group variation → evidence against H_0

Small F-value

Differences between group means are **small** relative to within-group variation → insufficient evidence to reject H_0

ANOVA Hypothesis Testing & Interpretation

Formal Hypotheses

Null Hypothesis *(H_0)*

All population group means are **equal**.

$$\mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$$

Alternative Hypothesis *(H_1)*

At least **one** population group mean is different from the others.

$$\exists\, i \neq j : \mu_i \neq \mu_j$$

Significance level: α = 0.05 is the standard threshold used in ANOVA.

Decision Rule

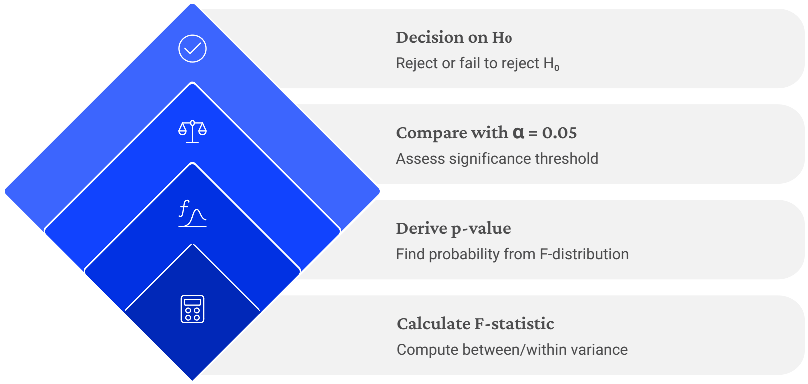
p-value < 0.05

Reject H_0 — a statistically significant difference exists between at least two group means.

p-value ≥ 0.05

Fail to reject H_0 — no statistically significant difference detected.

Decision Flowchart

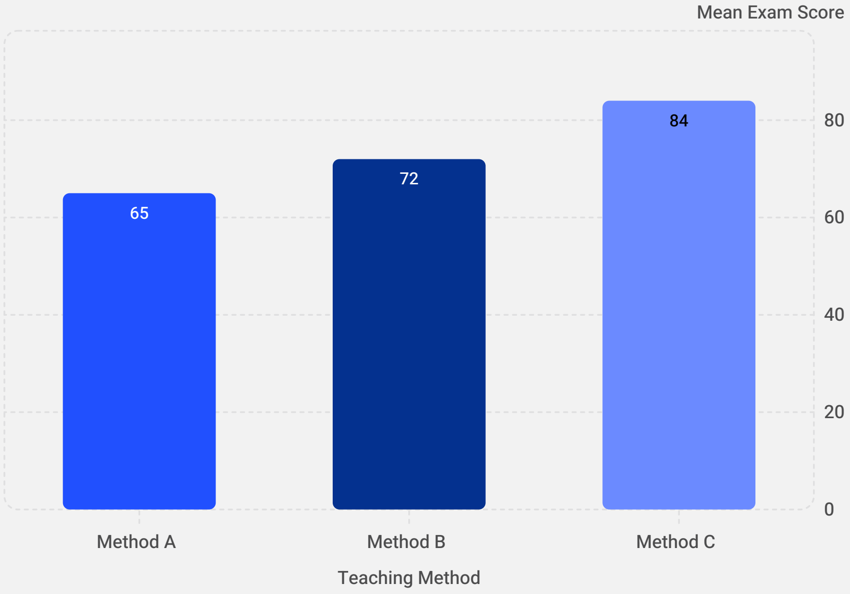


Important: ANOVA reveals *that* at least one group differs but does *not* identify *which* groups. Use **post-hoc tests** (e.g., Tukey's HSD) for pairwise comparisons.

ANOVA Example & Key Takeaways

Research Question: Do three teaching methods produce different average exam scores?

Data Summary & ANOVA Results



Variables in This Example

Categorical

Teaching Method — 3 independent groups (A, B, C)

Numerical

Exam Score — the outcome variable being compared across groups

Key Takeaways

1 **Multi-group comparison**

ANOVA compares means of **3 or more groups** in a single test

2 **Variation decomposition**

Separates **between-group** and **within-group** variation

3 **F & p-value**

Together determine **statistical significance** of group differences

4 **Essential EDA tool**

Core method for hypothesis testing involving categorical + numerical variables

Thank You

ANOVA – Analysis of Variance · Exploratory Data Analysis

Definition

Statistical test for significant differences across **3+ group means**

Mechanics

F-statistic = **Between-group** variance ÷ **Within-group** variance

Decision Rule

p-value < 0.05 → **Reject H_0**; significant difference confirmed

Next Steps

Use **post-hoc tests** (e.g., Tukey's HSD) to identify which groups differ

Questions & Discussion

ONE-WAY ANOVA — FULL NUMERICAL

Given: 3 teaching methods, each with 3 observations.
Method A: 10, 12, 14   Method B: 12, 14, 16   Method C: 15, 18, 21

TABLE

Method	X_1	X_2	X_3	$\bar{X}_i$
A	10	12	14	12
B	12	14	16	14
C	15	18	21	18

1) GROUP MEANS
$\bar{X}A = (10+12+14)/3 = 12$
$\bar{X}B = (12+14+16)/3 = 14$
$\bar{X}C = (15+18+21)/3 = 18$

2) GRAND MEAN
$\bar{X} = \Sigma X/N = (10+12+14+12+14+16+15+18+21)/9$
$= 132/9 = 14.67$

3) BETWEEN-GROUP SUM OF SQUARES
$SSB = \Sigma\, n_i(\bar{X}_i - \bar{X})^2$

Method	$\bar{X}_i$	$\bar{X}_i-\bar{X}$	$(\bar{X}_i-\bar{X})^2$	$n_i(\dots)$
A	12	−2.67	7.13	21.39
B	14	−0.67	0.45	1.35
C	18	3.33	11.09	33.27

$SSB \approx 21.39+1.35+33.27 = 56$

4) WITHIN-GROUP SUM OF SQUARES
Method A: $(10-12)^2+(12-12)^2+(14-12)^2 = 4+0+4 = 8$
Method B: $(12-14)^2+(14-14)^2+(16-14)^2 = 4+0+4 = 8$
Method C: $(15-18)^2+(18-18)^2+(21-18)^2 = 9+0+9 = 18$
$SSW = 8+8+18 = 34$

5) DEGREES OF FREEDOM & MEAN SQUARE
$dfB = k-1 = 3-1 = 2$ $MSB = SSB/dfB = 56/2 = 28$
$dfW = N-k = 9-3 = 6$ $MSW = SSW/dfW = 34/6 = 5.67$

6) F-STATISTIC
$F = MSB/MSW = 28/5.67 = 4.94$

7) DECISION AT $\alpha = 0.05$
$F_{critical}\ (df_1=2, df_2=6) \approx 5.14$
$F_{calculated} = 4.94 < 5.14 = F_{critical}$
$\therefore$ FAIL TO REJECT H_0

CONCLUSION: There is NO statistically significant difference between the three teaching methods at the 5% significance level.

Remember: $X_i \rightarrow \bar{X}_i \rightarrow$ Grand $\bar{X} \rightarrow SSB \rightarrow SSW \rightarrow MSB/MSW \rightarrow F \rightarrow$ Decision